

Median

Definition

The **Median** is a **measure of central tendency**. It is the single value that represents the **exact middle** of a dataset that has been arranged in ascending (smallest to largest) or descending (largest to smallest) order.

The median is also known as the **50th percentile** because it divides the dataset into two equal halves:

- 50% of the data values are **less than or equal to** the median.
- 50% of the data values are **greater than or equal to** the median.

The median is often preferred over the mean (average) when a dataset has **extreme values (outliers)**, because the median is not affected by them.



Formula For Both Odd And Even

The "formula" for the median is first about finding its **position** in an ordered dataset. Let N be the total number of observations (the count of all data points).

Step 1: Order the data from smallest to largest.

Step 2: Find the position of the median using the formula:

$$\text{Median Position} = \frac{N + 1}{2}$$

Case 1: N is Odd (e.g., $N = 5, 7, 9$)

- When N is odd, the formula $\frac{N+1}{2}$ gives a whole number.

- **Formula:** The median is the value at this *exact position*.
- **Example:** If $N = 7$, the position is $\frac{7+1}{2} = 4$. The median is the **4th value** in the ordered list.

Case 2: N is Even (e.g., $N = 6, 8, 10$)

- When N is even, the formula $\frac{N+1}{2}$ gives a decimal (e.g., 3.5, 5.5).
- A position like "3.5" means the median lies *between* the 3rd and 4th values.
- **Formula:** The median is the **average of the two middle values** (the values on either side of the decimal position).
- **Example:** If $N = 6$, the position is $\frac{6+1}{2} = 3.5$. The median is the **average of the 3rd and 4th values**.

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Examples

Ungrouped Data

This is a raw list of numbers.

Example 1: N is Odd

- **Data:** 19, 12, 16, 25, 14
- **Step 1 (Order):** 12, 14, 16, 19, 25
- **Step 2 (N):** $N = 5$ (Odd)
- **Step 3 (Position):** $\frac{5+1}{2} = \frac{6}{2} = 3$ rd position.
- **Step 4 (Median):** The 3rd value in the ordered list is **16**.

Example 2: N is Even

- **Data:** 30, 10, 20, 50, 40, 60
- **Step 1 (Order):** 10, 20, 30, 40, 50, 60
- **Step 2 (N):** $N = 6$ (Even)
- **Step 3 (Position):** $\frac{6+1}{2} = \frac{7}{2} = 3.5$ th position.

- **Step 4 (Median):** The position 3.5 is between the 3rd and 4th values.
 - 3rd value = 30
 - 4th value = 40
 - Median = $\frac{30+40}{2} = \frac{70}{2} = \mathbf{35}$

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Discrete Frequency Distribution

This is when data is presented in a table showing each value (x) and how many times it occurs (frequency, f).

Procedure:

1. Calculate the **Cumulative Frequency (cf)**. This is a running total of the frequencies.
2. Find the total frequency, $N = \sum f$.
3. Find the median position using $\frac{N+1}{2}$.
4. Locate the cf value that is **just greater than or equal to** the median position.
5. The median is the **x value** (the data value, not the frequency) corresponding to that cf .

Example:

x (Marks)	f (Students)	cf (Cumulative Freq.)
10	2	2
20	5	$2 + 5 = 7$
30	8	$7 + 8 = 15$
40	3	$15 + 3 = 18$
50	2	$18 + 2 = 20$

- **Step 2 (N):** $N = \sum f = 20$ (Even)
- **Step 3 (Position):** $\frac{20+1}{2} = \frac{21}{2} = 10.5$ th position.
- **Step 4 (Locate cf):** We need to find the cf just greater than or equal to 10.5.
 - cf of 2 is too small.

- cf of 7 is too small.
- cf of 15 is the first one greater than 10.5.
- **Step 5 (Median):** The x value corresponding to the cf of 15 is 30.
 - (This means the 8th, 9th, 10th, 11th, 12th, 13th, 14th, and 15th students all scored 30. Since we are looking for the 10.5th value, it falls in this group.)

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Continuous Frequency Distribution

This is when data is grouped into class intervals (e.g., 10 – 20, 20 – 30). We first find the **Median Class** and then use a formula to estimate the median value within that class.

Procedure:

1. Calculate the **Cumulative Frequency (cf)**.
2. Find the total frequency, $N = \sum f$.
3. Find the median **class location** using $\frac{N}{2}$. (Note: We use $\frac{N}{2}$ here, not $\frac{N+1}{2}$, as per standard convention for continuous data).
4. Find the **Median Class**: This is the class interval whose cf is **just greater than or equal to** the $\frac{N}{2}$ value.
5. Apply the Median Formula:

$$\text{Median} = L + \frac{\left(\frac{N}{2} - cf\right)}{f} \times i$$

Where:

- L = **Lower limit** of the Median Class.
- N = **Total frequency** ($\sum f$).
- cf = **Cumulative frequency of the class preceding (before) the Median Class**.
- f = **Frequency** of the **Median Class** itself.
- i = **Class width** (Upper limit - Lower limit) of the Median Class.

Example:

Class Interval (Marks)	f (Students)	cf (Cumulative Freq.)
0 – 10	5	5
10 – 20	10	$5 + 10 = 15$
20 – 30	20	$15 + 20 = 35$
30 – 40	8	$35 + 8 = 43$
40 – 50	7	$43 + 7 = 50$

- **Step 2 (N):** $N = 50$
- **Step 3 (Location):** $\frac{N}{2} = \frac{50}{2} = 25$ th position.
- **Step 4 (Median Class):** We look for the cf just greater than or equal to 25.
 - cf of 5 is too small.
 - cf of 15 is too small.
 - cf of **35** is the first one greater than 25.
 - Therefore, the **Median Class** is **20 – 30**.
- **Step 5 (Apply Formula):**
 - $L = 20$ (Lower limit of 20 – 30)
 - $N = 50$
 - $cf = 15$ (The cf of the class *before* 20 – 30)
 - $f = 20$ (The frequency of the 20 – 30 class)
 - $i = 10$ (30 – 20)

$$\text{Median} = 20 + \frac{\left(\frac{50}{2} - 15\right)}{20} \times 10$$

$$\text{Median} = 20 + \frac{(25 - 15)}{20} \times 10$$

$$\text{Median} = 20 + \frac{10}{20} \times 10$$

$$\text{Median} = 20 + 0.5 \times 10$$

$$\text{Median} = 20 + 5$$

$$\text{Median} = \mathbf{25}$$

The median mark is 25. This value makes sense as it falls within our 20 – 30 median class.